

COOLING TOWER OPERATIONS AND MAINTENANCE MANUAL

CT-1 and CT-2 - Induced Draft Counterflow

1. PURPOSE

Maintain heat rejection capacity for chillers and process cooling. Predictive focus: fan vibration, gearbox oil, basin water quality, and fill condition.

2. OPERATING PARAMETERS

Approach temperature: design 5-7 F above wet bulb

Basin water temperature: 29-35 C typical summer

Fan motor current: within 10% of baseline

Blowdown rate: maintain conductivity per water treatment sheet

3. DAILY OPERATOR ROUNDS

- Inspect fan operation and unusual vibration
- Check drift eliminator carryover (excess water loss)
- Verify chemical feed pumps operating
- Record basin level and makeup water meter

4. WEEKLY MAINTENANCE

- Clean suction strainer
- Inspect belt tension on fan drive (if belt-driven)
- Check gearbox oil level and leaks
- Sample basin water for Legionella protocol per health code

5. MONTHLY TASKS

- Lubricate fan bearings per LUB-G2-LC chart
- Inspect fill for scaling or biological growth
- Thermography on fan motor and gearbox

6. PREDICTIVE TRIGGERS

Schedule fill section replacement when:

- Approach temp degrades 15% at same load and wet bulb
- Visible fill collapse or heavy scale
- Fan vibration exceeds 7.1 mm/s for 3 consecutive readings

7. WATER TREATMENT

Maintain biocide, scale inhibitor, and corrosion inhibitor per vendor. Over-blowdown wastes water; under-blowdown scales fill.

8. SEASONAL

Spring startup: disinfect basin, inspect louvers, test vibration switches.

Fall shutdown (if applicable): drain basin, protect motors from freeze.

9. SAFETY

Lockout fans before entering cell. Fall protection required on elevated walks. Legionella risk - no pressure washing without mist control PPE.

10. SPARES

Fan belts, gearbox oil GL-220, fill sections (lead time 8 weeks), level switch floats.